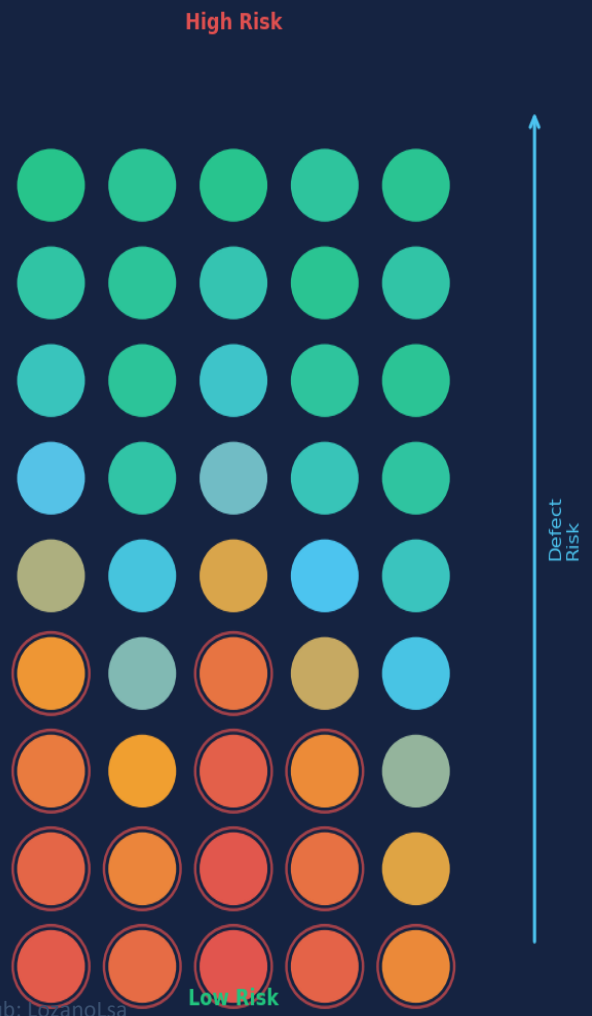
MACHINE LEARNING · QUALITY INSPECTION · PREDIC

Visual Defects Are Not Random.

What if surface defects were not inevitable… but predictable?

Naive Bayes · Risk Scoring · Defect Prevention



1,500
Painted Parts

10
Features

76.2%
Accuracy

0.629
ROC-AUC

The Real Problem

A surface defect is not just a cosmetic issue — it's a failure you can't hide.

Quality Cost

Scrap, rework, and repaint consume labor and cycle time.

Customer Visibility

A painted trim piece with a scratch goes directly to the customer.

Late Discovery

Defects appear at final inspection but originate upstream.

The defect emerges at inspection — but the root cause lives in injection, storage, and handling. That gap is where ML operates.

In manufacturing, every defective part has a history. The question is whether we're reading it.

From Events to Probabilities

Old Thinking

Defects are random events:

"It happened / it didn't."

Inspect at the end.
Reject what fails.
No prevention. No foresight.
Just sorting after the damage.

New Thinking

Defects are probabilistic outcomes:

"How likely is a defect — before painting?"

Score each part before it enters the cabin.
High-risk parts get extra attention.
Low-risk parts flow through.
Defect risk becomes a process input.

The right question is not 'Did it pass?' — it's 'How likely is a defect, and why?'

The Data Behind the Problem

This dataset spans four process stages — each contributing its own defect risk:

INJECTION	**Regrind %, resin temperature, cooling time** *Injection molding stage — material and process conditions*
PAINT CABIN	**Paint viscosity, film thickness, booth humidity** *Paint application stage — coating quality parameters*
LOGISTICS	**Storage time, handlings, container type, protection** *Internal logistics — the dominant defect risk stage*
INSPECTION	**surface_defect (0 = pass, 1 = defect)** *Final visual inspection — the binary target variable*

Key insight: defects originate in logistics, not in painting. The model quantifies what intuition suspected.

In a real environment, this data lives in four different systems — assembling it is half the work.

Why Naive Bayes?

A probabilistic model that reasons like an engineer: assign a risk score based on known conditions.

No Scaling Needed

GaussianNB estimates mean and variance per class directly — no StandardScaler required. Simpler, faster, production-ready.

Probabilistic Output

Returns defect probability (0–1), not just pass/fail. Enables risk ranking and threshold adjustment at the operator's discretion.

Pipeline-Ready

sklearn Pipeline bundles preprocessing and classification. The simulator in Section 10 is portable with a single predict_proba() call.

Effect Size Auditable

Cohen's d and Mutual Information reveal which features actually separate defect from pass — no black box, full auditability.

Simplicity is a feature — a model your quality team can explain is a model they'll actually use.

Model Results

76.2%	0.629	0.101	21.9%
Accuracy	ROC-AUC	F1 (Defect)	Defect Rate

Top Drivers — Cohen's d & Mutual Information

pre_paint_storage_hrs	d = 0.189
num_handlings	d = 0.178
part_protection (MI)	Dominant
container_type (MI)	Secondary
resin_temp_c	d = 0.063
regrind_pct	d = 0.017

Confusion Matrix (450 test parts)

	Pred: Pass	Pred: Defect
Actual: Pass	337 (TN)	15 (FP)
Actual: Defect	92 (FN)	6 (TP)

Low recall at 0.5 threshold — the model's real value is the probability score, not the binary verdict.

AUC > 0.5 confirms real signal. Recall improves dramatically with threshold tuning.

From Insight to Action

Storage Time d=0.189	→ Cap pre-paint waiting at 4 hours. Parts stored longer show measurably higher defect rates — contamination and static accumulate.
Handlings d=0.178	→ Reduce repositioning steps in internal logistics. Each extra handling = +0.3pp defect risk on average.
Unprotected 28.8% rate	→ Make part protection mandatory for all batches entering the paint queue. Protected parts show 16.8% vs 28.8% defect rate.
Cardboard 28.9% rate	→ Migrate from cardboard pallets to plastic boxes. Plastic_box containers show 15.6% vs 28.9% for cardboard.
Injection params	→ Resin temp, cooling time, regrind have near-zero Cohen's d — the defect is not born at injection. Focus upstream effort on logistics.

The model doesn't inspect parts — it tells you where to act before they enter the paint cabin.

What-If Simulator — Risk Scorecard

Same model. Three scenarios. The third one isolates the value of a single operational decision.

Scenario A — Low Risk		Scenario B — High Risk		Scenario C — Intervention	
Storage	1 hr	Storage	18 hrs	Storage	18 hrs
Handlings	2	Handlings	6	Handlings	6
Container	plastic_box	Container	metal_rack	Container	metal_rack
Protection	protected	Protection	unprotected	Protection	**protected ✓**
Temp	235°C	Temp	248°C	Temp	248°C
4.6% defect probability		**97.6% defect probability**		**95.5% defect probability**	

Scenario C insight: adding protection alone drops the risk by only 2.1pp. The dominant drivers are storage time and handling count — not packaging.

Probability is a guide, not a verdict. The simulator turns model output into pre-paint operational decisions.

This project is not about finding defects.

It is about preventing them.

Naive Bayes does not replace inspection —
it shifts the decision upstream, before the damage is done.

Feel free to download the full project: dataset, notebook, and simulator.

Change the thresholds. Adjust the scenarios. Challenge the assumptions.

Because quality has no finish line.